Saakash Sathyan

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Education

University of California, Irvine — Expected June 2027

B.S. Aerospace Engineering

- Relevant Coursework: Dynamics, Fluid Dynamics I-III, Thermodynamics, Astronautics, Circuits, Materials Science, Aerodynamics, Mechanics of Structures, Mechanical Systems Lab
- Relevant Organizations: American Institute of Aeronautics and Astronautics (AIAA), Students for the Exploration and Development of Space (SEDS)

Projects

AI-Integrated Companion Robot | *Independent Project* — Jul 2026 – Present

- Collaborating with an industry mentor (System Architect, Google) and two peers to design an AI-driven companion robot
- Developed initial concept sketches and block-component CAD models defining system layout and basic dimensioning
- Built bills of materials with component trade-offs across compute, sensing, and actuation subsystems
- Researching split compute architecture utilizing Raspberry Pi for AI processing and ESP32 for motion control
- Evaluating sensor and camera configurations for obstacle avoidance, edge detection, and voice interaction

SEDS @ UCI Autonomous Mars Rover | *Fabrication Subteam* — Nov 2025 – Present

- Designed autonomous rover to traverse uneven terrain and retrieve a rock sample without human input
- Modeled full chassis assembly in SolidWorks including dual-deck frame, 4 motor mounts, and wheel hubs
- Integrated 3 subassemblies (turning mechanism, 2-DOF arm, claw) into the full rover assembly
- Created system-level concept sketches defining rover architecture and component relationships across 3 subteams
- Fabricated arm, claw, and turning mechanism components using Orca Slicer and Bambu X1C 3D printers

UCI MAE 106 Autonomous Pneumatic Robot | *Mechanical Coordinator* — Mar 2026 – Jun 2026

- Designed full chassis assembly in SolidWorks including frame, wheel mounts, and steering linkage
- Engineered pneumatic propulsion system harnessing a single cylinder powered by a compressed air tire
- Designed steering mechanism actuated by servo motor to enable autonomous directional control
- Fabricated components using band saw, drill press, laser cutter, and 3D printer with PLA filament
- Robot competed autonomously using bearing-based orientation and wheel-rotation odometry for navigation

Leadership & Activities

Internal Affairs Director — Jun 2026 – Present

AIAA @ UCI — *Irvine, CA*

- Organize and facilitate general chapter meetings and on-campus events
- Manage event logistics and coordinate scheduling for chapter programming
- Lead recruitment efforts to grow and develop the internal affairs team
- Coordinate member engagement initiatives across a 100+ member organization

Financial Director — Nov 2025 – Jun 2026

AIAA @ UCI — *Irvine, CA*

- Oversee budget planning and financial reporting for a ~$10,000 annual budget
- Manage financial operations supporting a 100+ member student engineering organization
- Personally managed 2 fundraising events from planning through execution and financial reconciliation
- Coordinate sponsorship acquisition and develop proposals for aerospace industry partners

Certifications & Training

UAV/Drone Design and Analysis Training Program — May 2026 - Jun 2026

Skyline Space — *India - Remote*

- Completed coursework in UAV aerodynamics, flight dynamics, stability analysis, and autonomous systems
- Utilized SolidWorks and ANSYS software to model drone components and perform CFD simulations

Technical Skills

Programming: MATLAB, Python, C++, RobotC
CAD & Design: SolidWorks, AutoCAD, Fusion 360, OnShape, Inventor, ANSYS, Parametric Modeling, Orca Slicer
Engineering: Mechanical Design, CAD Modeling, System Integration, Component Assembly, Testing, Aerodynamic Analysis, CFD Simulation, Technical Documentation, Design Review
Fabrication & Hardware: Hand & Power Tools, Band Saw, Table Saw, Belt Sander, FDM 3D Printing, Soldering, Oscilloscope, Function Generator, Digital Calipers
Productivity: Microsoft Office, Google Suite, Adobe Photoshop